

Weather Trend Forecasting

PM Accelerator Mission

By making industry-leading tools and education available to individuals from all backgrounds, PM Accelerator helps level the playing field for future professionals in Product Management, Data Science, Artificial Intelligence, and Technology. Through mentorship, networking, and project-based learning, PM Accelerator provides opportunities for individuals to gain practical experience and develop industry-relevant skills.

Objective

The objective of this project is to analyze global weather data and forecast future weather trends using machine learning techniques. The dataset contains daily weather observations from locations around the world, including temperature, humidity, precipitation, wind speed, atmospheric pressure, air quality measurements, and geographical information.

Basic Assessment

Data Cleaning and Preprocessing

Before performing the analysis, the dataset was cleaned and prepared.

The following preprocessing steps were completed:

- Checked for missing values and handled them appropriately.
- Removed duplicate records.
- Converted the `last_updated` column into datetime format.
- Created additional date-related features such as year, month, and day.
- Detected outliers using the Interquartile Range (IQR) method.
- Normalized numerical variables using `StandardScaler`.

Observation

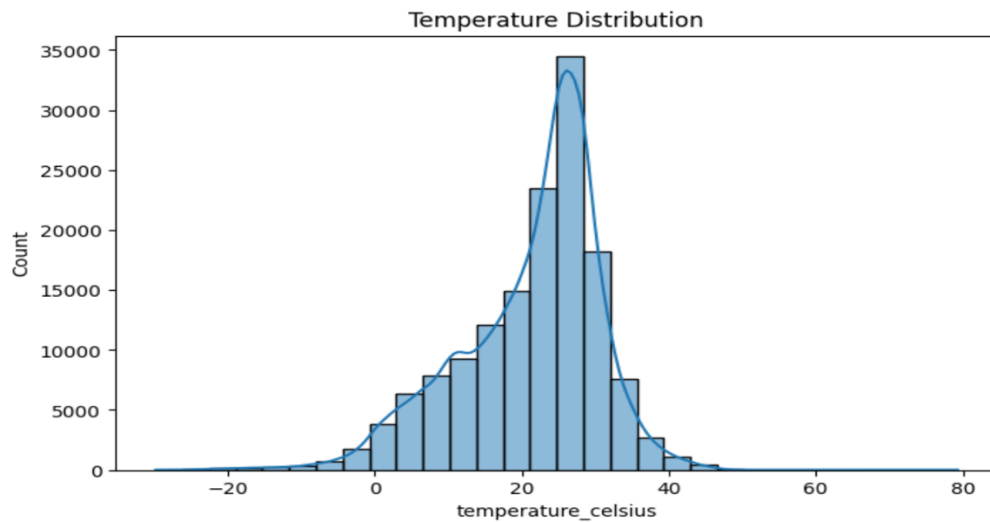
The dataset was relatively clean with very few quality issues. Converting the date column allowed the data to be used for time-series forecasting and trend analysis.

Exploratory Data Analysis (EDA)

EDA was performed to understand the distribution of weather variables and identify relationships between them.

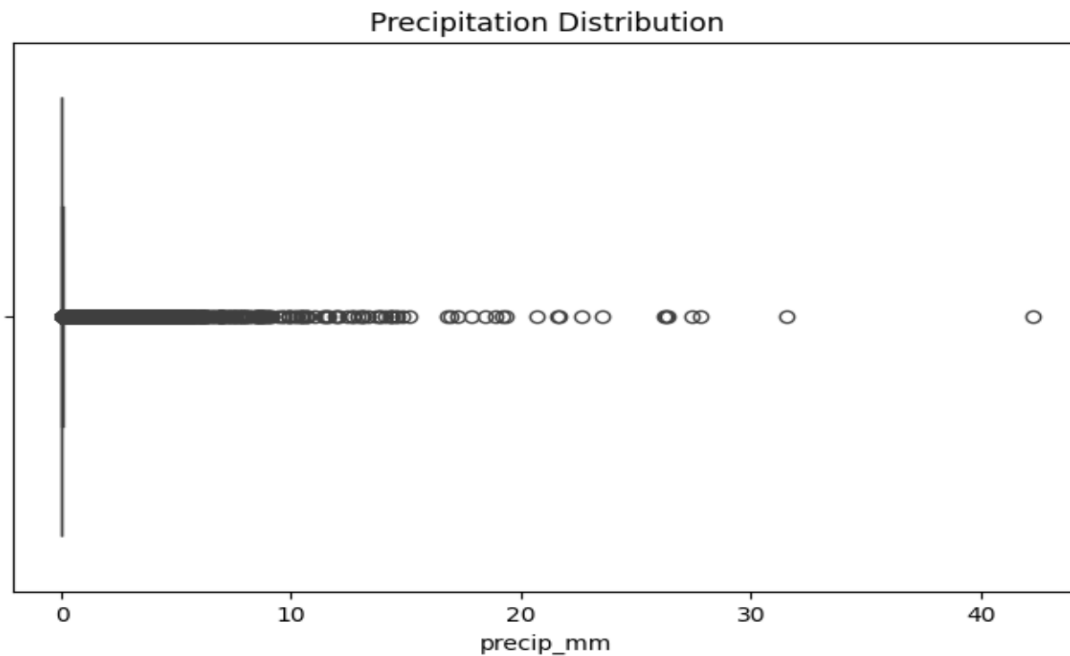
Temperature Distribution

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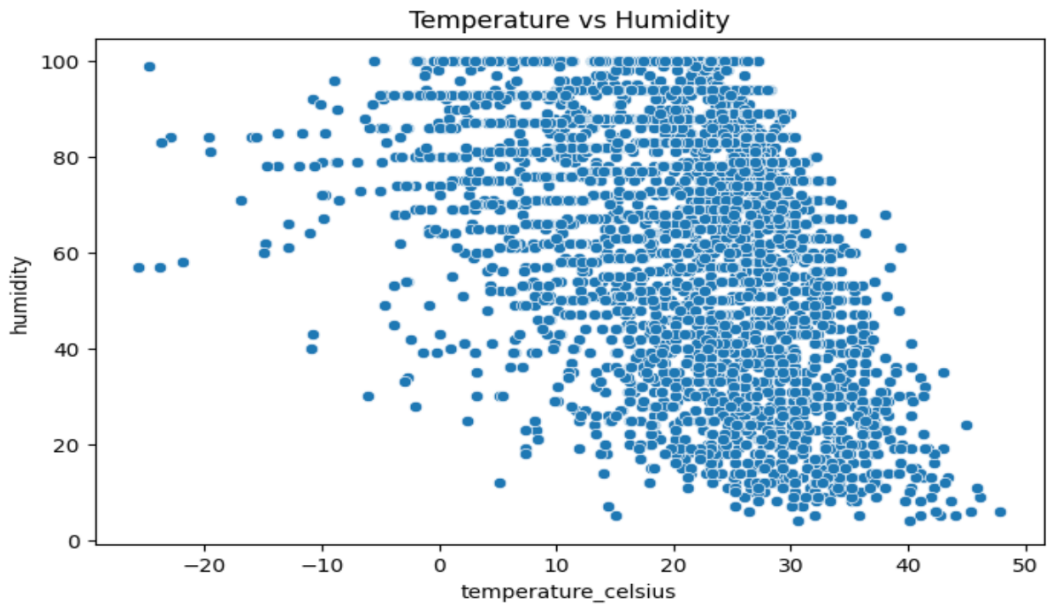
Observation: Most temperature observations were concentrated within moderate ranges, while extreme temperatures occurred less frequently.

Precipitation Distribution



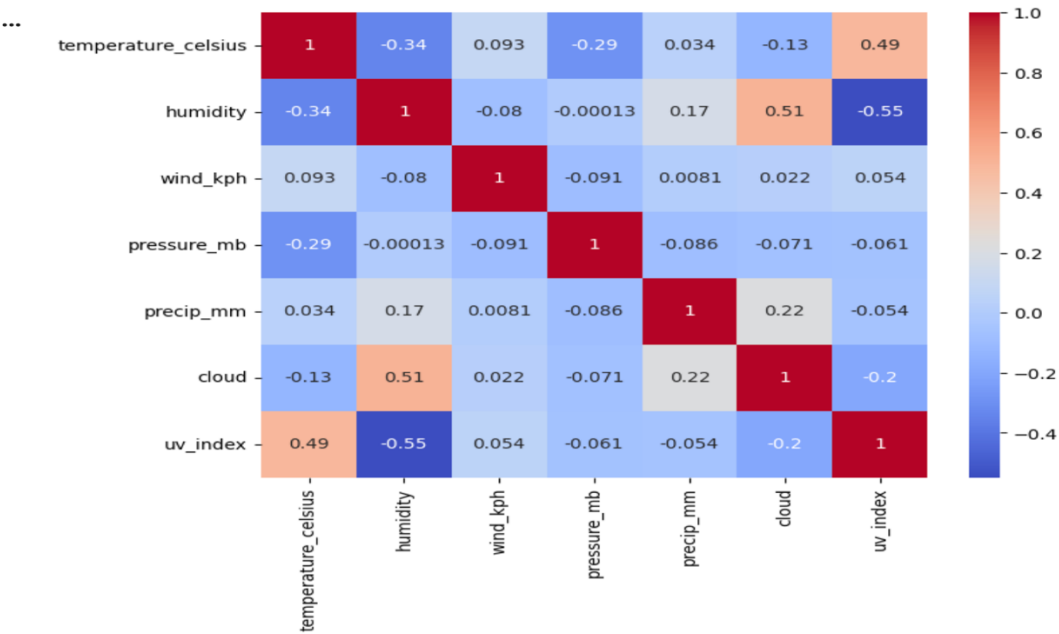
Observation: The precipitation data was highly skewed. Most observations recorded little or no rainfall, while a small number of records represented heavy precipitation events.

Temperature vs Humidity Analysis



Observation: The scatter plot revealed varying relationships between temperature and humidity across different weather conditions. Humidity appears to be an important factor influencing temperature behavior and was later identified as one of the important features for prediction.

Correlation Analysis



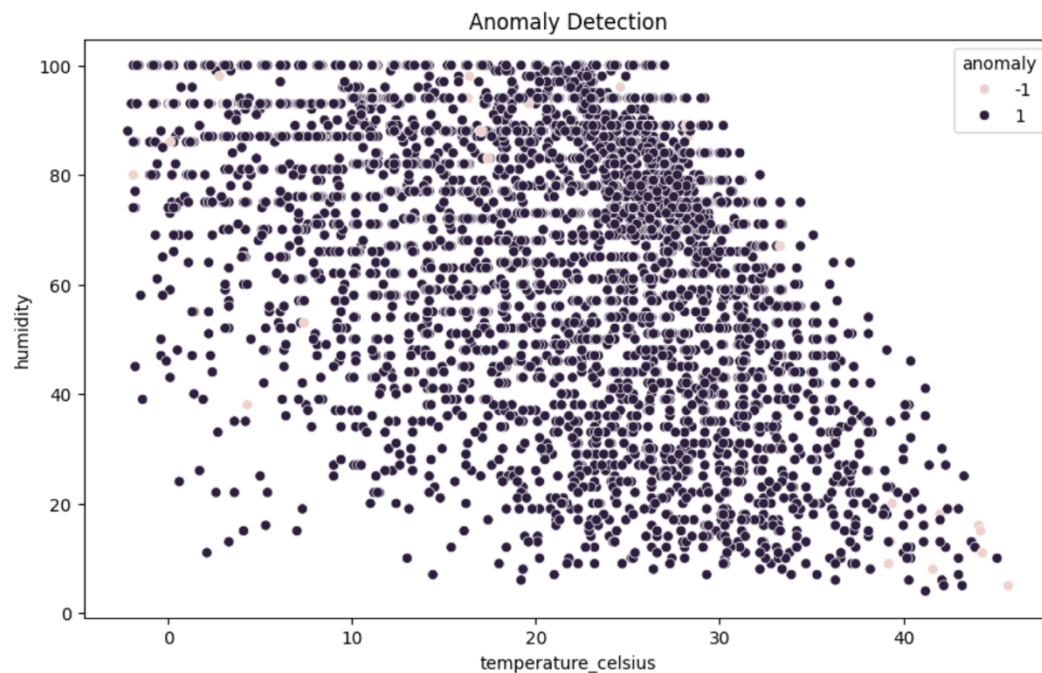
Observation: Temperature, humidity, pressure, cloud coverage, and wind speed showed meaningful relationships that may be useful for weather forecasting models.

Advanced Assessment

Advanced EDA – Anomaly Detection

The Isolation Forest algorithm was used to identify unusual weather observations.

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Observation

The anomaly detection model identified weather records that significantly differed from typical patterns. These observations may represent extreme weather events or unusual environmental conditions.

Forecasting with Multiple Models

To improve forecasting accuracy, multiple machine learning models were developed and compared.

The models included:

- ARIMA
- Random Forest Regressor
- Ensemble Model

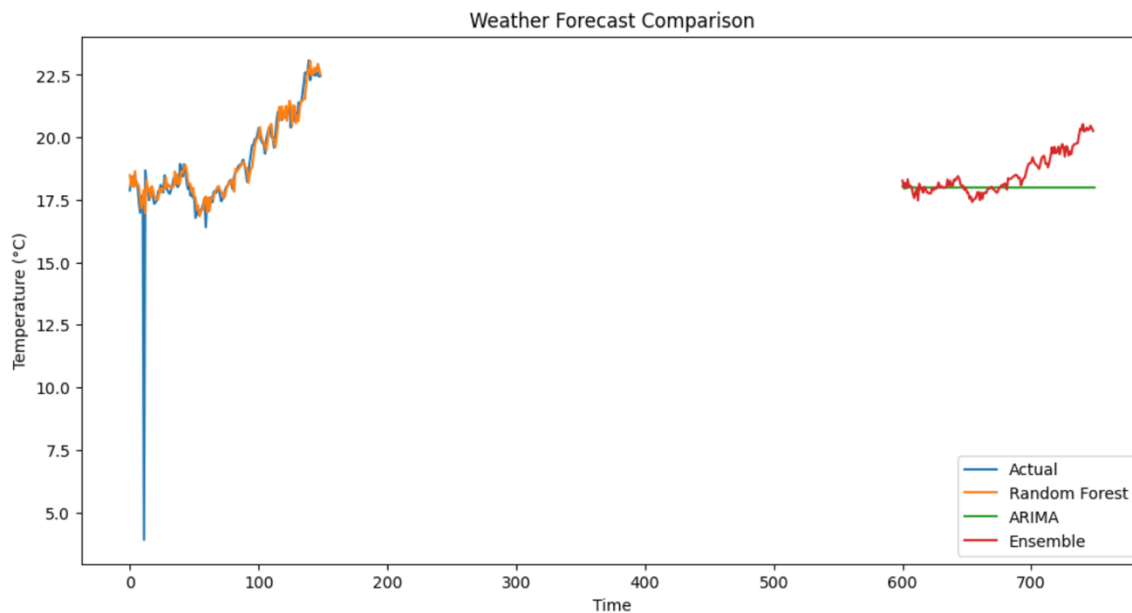
The models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Model Comparison

	Model	MAE	RMSE	R2 Score
0	ARIMA	1.470184	2.289696	-0.229027
1	Random Forest	0.383599	1.210976	0.658268
2	Ensemble	0.853088	1.554445	0.436925

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Observation

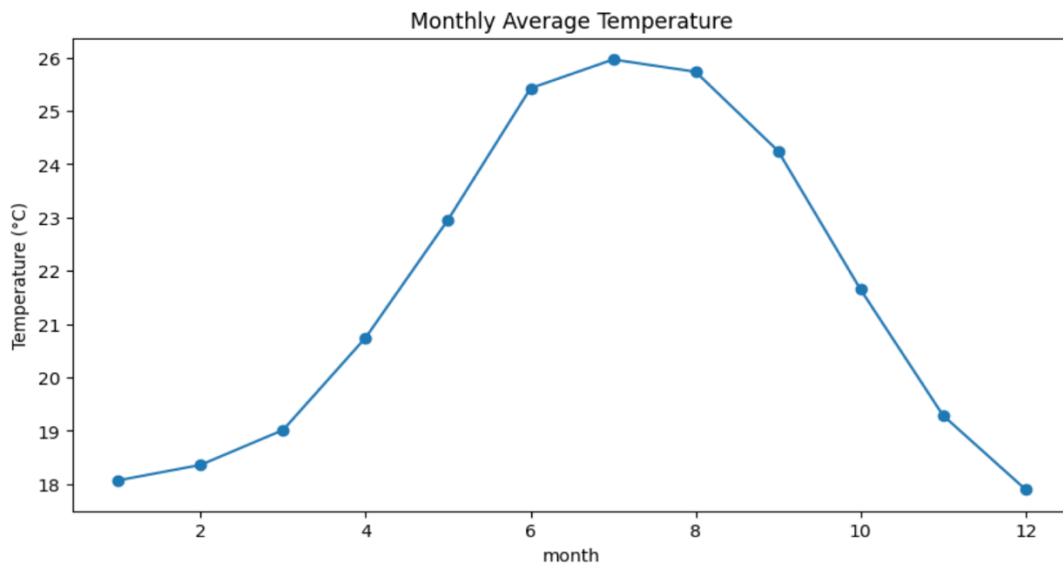
Random Forest achieved the best predictive performance with the lowest MAE and RMSE and the highest R^2 score. ARIMA produced the most stable forecast but was less effective at capturing temperature variations. The Ensemble model provided a balance between stability and trend-following behavior

Unique Analyses

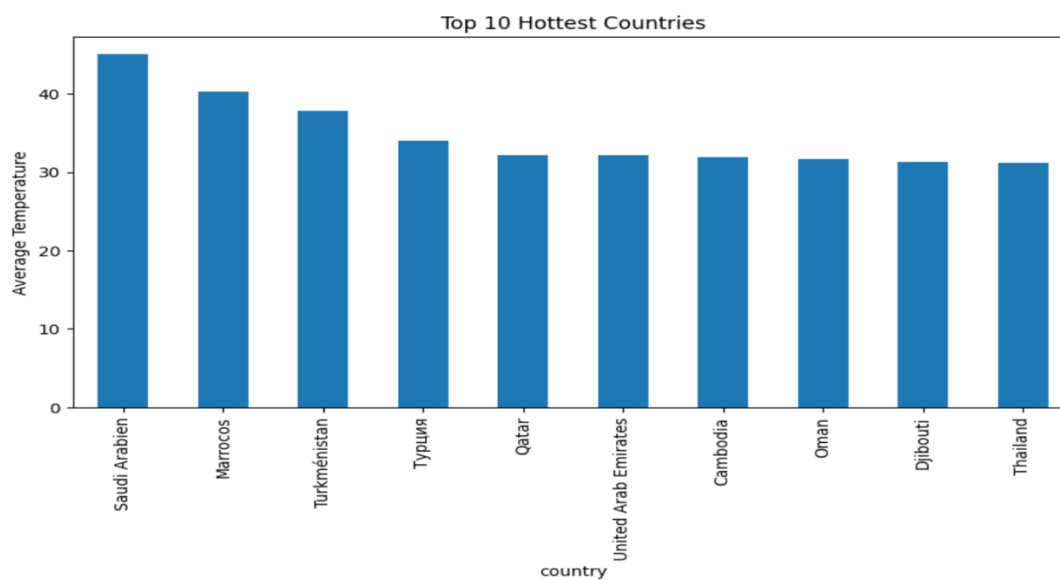
Climate Analysis

Climate patterns were analyzed by examining temperature trends across different months and countries.

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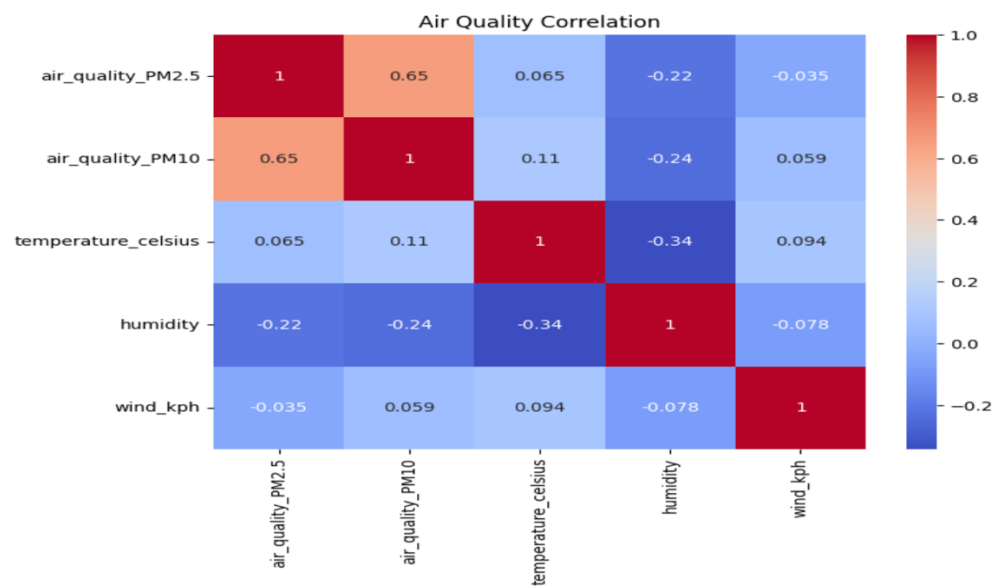


Observation

The analysis revealed noticeable seasonal patterns in temperature data. Countries located in warmer climates generally recorded higher average temperatures compared to countries in cooler regions.

Environmental Impact Analysis

Weather variables were analyzed alongside air quality indicators.

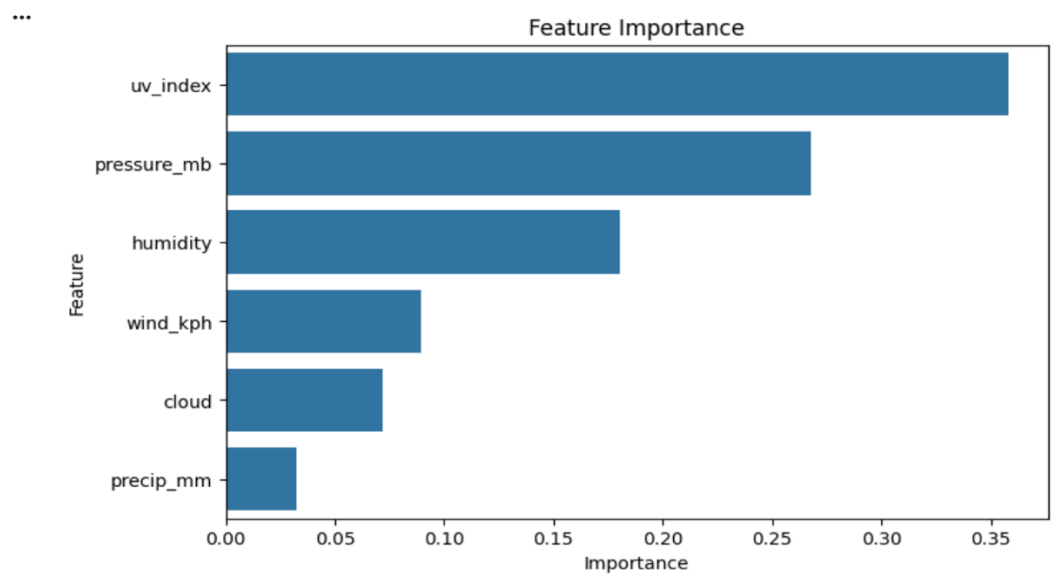


Observation

Air quality measurements such as PM2.5 and PM10 showed relationships with weather conditions. Temperature, humidity, and wind speed appeared to influence pollutant levels in different regions.

Feature Importance Analysis

Feature importance was evaluated using the Random Forest model.

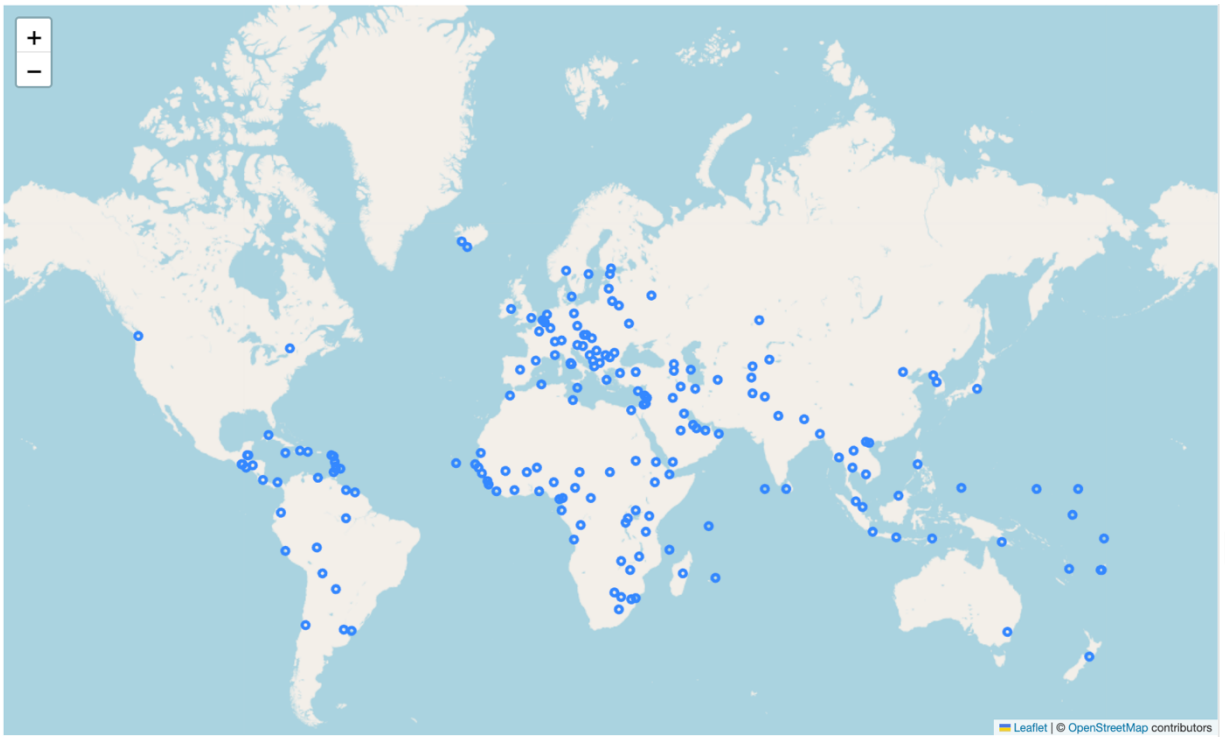


Observation

The analysis showed that humidity, pressure, UV index, and cloud coverage were among the most influential variables for temperature prediction.

Spatial Analysis

Geographical weather patterns were visualized using latitude and longitude information.

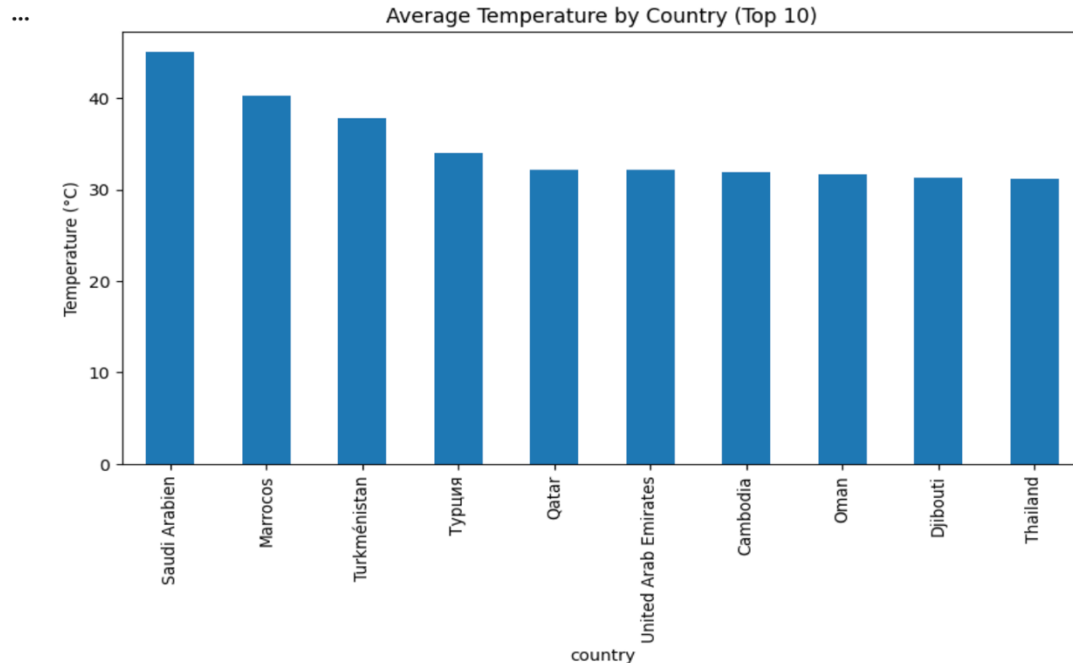


Observation

The spatial analysis showed that weather conditions vary significantly across geographic regions. Locations closer to the equator generally experienced higher temperatures than locations farther away.

Geographical Patterns

Weather characteristics were compared across countries and regions.



Observation

The analysis revealed clear geographical differences in temperature, humidity, precipitation, and wind speed. These variations highlight the influence of location and climate on weather behavior.

Conclusion

This project successfully analyzed global weather data using both basic and advanced data science techniques. Data preprocessing, exploratory analysis, anomaly detection, climate analysis, environmental impact assessment, feature importance evaluation, spatial analysis, and forecasting were performed.

The forecasting models demonstrated the ability to predict temperature trends using historical weather observations. In addition, the advanced analyses provided insights into climate variation, environmental conditions, and geographical weather patterns around the world.

Overall, the project demonstrates how data science and machine learning techniques can be applied to understand weather behavior and support future forecasting efforts.